

Question 1:How does a VPC ensure secure and isolated networking in AWS?

Answer Points:

- VPC lets you create your own private cloud network inside AWS.
- You control subnets, IP address ranges, and routing rules.
- Provides Security Groups and NACLs for traffic filtering.
- Supports public and private subnets to separate internet-facing and internal systems.

Scenario Example : Think of a VPC like building a housing layout — you decide which houses face the main road (public subnet) and which stay inside a gated community (private subnet).

Question 2:What are the common use cases for EC2 in AWS?

Answer Points:

- Hosting websites and web applications.
- Running backend services, APIs, or application servers.
- Performing batch processing or data analysis workloads.
- Hosting Windows or Linux environments for administration or testing.

Scenario Example :A small company uses an EC2 Windows instance like a remote office computer where the team logs in to manage internal software.

Question 3:What is Continuous Deployment and how does it help DevOps teams?

Answer Points:

- Automatically deploys each successful build directly into production.
- Removes manual approval steps, speeding up delivery.
- Ensures consistent, reliable deployments across environments.
- Reduces human error by automating release processes.

Scenario Example: It's like an automatic conveyor belt in a factory — every finished product moves straight to the shelf without waiting for someone to manually push it.

Question 4: Why are Docker containers widely used in DevOps?

Answer Points:

- They package applications with all required dependencies.
- Ensure consistent execution across development, test, and production.
- Start quickly and use fewer resources compared to full VMs.
- Work with orchestration platforms like Kubernetes and ECS.

Scenario Example: Docker containers are like delivery lunch boxes — no matter where you open them, the food is arranged exactly the same.